



BUSINESS ANALYSIS REPORT

client
Popular Auto Spare and A/C Parts Co. W.L.L



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VENDOR & PURCHASE
PROCESS FLOW

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PREFACE

This document describes the workflow and functional processes related to the Purchase and Vendor Management System. It has been prepared based on the discussions held with the management and operational teams, with the objective of clearly defining the purchase procedures, data flow, approval hierarchy, and standard operating practices.

The document outlines the end-to-end process, starting from branch requirements and stock shortages, through the purchase request process, vendor quotation management, purchase order generation, verification procedures, and final stock updates. It also explains the roles and responsibilities of each department involved, including the Branches, Main Store, Purchase Department, and Management.

This document primarily concentrates on the workflow of the Purchase Department and its integration with the Main Store and Branch operations. While this document defines the purchase-related processes and associated data flow in detail, the complete branch-level operational workflow is described separately in the next document Doc-003 v-1. The objective here is to clearly establish how purchase activities are initiated, processed, approved, executed, and recorded within the system.

In addition to the functional workflow, this document includes key management control points, vendor evaluation logic, approval stages, audit tracking mechanisms, and structured data flow processes required to ensure accuracy, transparency, compliance, and accountability in procurement activities.

This preface serves as an introduction to the overall Purchase and Vendor Management framework. The document should be used as a reference for system development, implementation planning, operational execution, and standardization across departments. The contents of this document are based on the requirements and discussions conducted during the analysis phase and are intended to guide both technical development and operational usage of the system.





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1. Vendor Master Management

Popular Auto Spare and A/C Parts Co. operates as a multi-country, multi-branch enterprise; however, the fundamental operating principle is that all procurement activities are centrally controlled from a designated headquarters location. This centralized purchasing model ensures uniform pricing strategy, supplier governance, negotiation control, and compliance consistency across all branches and countries.

Vendors from different countries will be registered within a unified global Vendor Master. Each supplier record will capture country-specific regulatory, tax, currency, and banking details while remaining governed under centralized approval authority. This structure allows the company to maintain a single source of truth for supplier data while supporting international procurement operations.

Vendor master records include the following key parameters:

- Vendor name
- Vendor code
- Legal entity name
- Contact person details
- Phone and email information
- Registered address
- Country and operating region
- Tax/VAT registration details
- Payment terms
- Currency preference
- Bank and remittance details
- Supplier category (OEM / Aftermarket / Local / International)
- Brand association (if applicable)
- Credit limit and risk classification
- Status (Active / On Hold / Inactive)

2. Vendor Communication Framework – Current vs Proposed System

Current Scenario

In the existing operational environment, communication between the purchase team and vendors is handled through external third-party applications such as email and messaging platforms including WeChat and WhatsApp. While these tools enable quick interaction, they introduce several operational limitations:



- Communication is fragmented across multiple platforms
- No centralized record of vendor conversations
- Difficulty in tracking quotation history and negotiations
- Loss of institutional knowledge when staff change
- No audit trail for compliance review
- Dependency on personal accounts or unmanaged tools
- Risk of data leakage and confidentiality exposure

As a result, vendor communication lacks governance, traceability, and long-term organizational memory.

Proposed ERP-Integrated Vendor Communication Model

The proposed ERP introduces a secure, system-controlled vendor communication framework that replaces or supplements third-party tools with a structured digital channel.

Key Concept:

For request for quotation, the system generates a unique secure vendor link. This link acts as a controlled communication portal between the supplier and the purchase department.

Once the user sends the RFQ link to the vendor for pricing, the vendor can access the link and update the price and availability details for the specified products.

After the vendor submits the quotation, the CTO has the authority to lock the RFQ link. Once the link is locked, the vendor can still log in and view the submitted quotation details, but they will not be able to modify or update any values.

If any revisions are required, the CTO can unlock the RFQ link, allowing the vendor to update and resubmit the necessary changes. This mechanism ensures controlled quotation management while maintaining flexibility when corrections are needed.

Core Features

1. Unique Transaction Link

- Each supplier receives a dedicated secure link tied to a specific transaction
- Links are time-stamped and system-tracked
- Access is restricted to authorized vendor representatives

2. Integrated Chat Interface

- A built-in chat interface is embedded inside the link
- Suppliers and purchase staff communicate within the ERP ecosystem
- No need to switch to external messaging platforms

3. Permanent Communication History

- All chat messages are stored automatically



- Linked directly to the transaction record
- Searchable for future reference and dispute resolution
- Visible to authorized management users

4. Conversation Continuity

- Each time the vendor opens the link, the full chat history is displayed
- Ensures transparency and continuity
- Eliminates repeated explanations or lost context

Implementation Options

Option 1 – Full ERP Communication Model

- Replace third-party apps completely
- All vendor communication happens inside ERP
- Maximum governance and audit control

Option 2 – Hybrid Communication Model

- ERP communication becomes the official record
- External apps may still be used for informal speed
- Critical communication logged inside ERP
- Gradual transition strategy

3. Automated Low-Stock Reporting

Current Scenario

At present, low-stock reporting from branches to the centralized purchase department is handled manually using Excel templates.

Branch managers or sales supervisors prepare a daily spreadsheet containing:

- OEM number
- Brand
- Model
- Item description
- Available quantity
- Required quantity

The file is then emailed to the purchase department.

This process creates operational pressure and inefficiency:

- Sales staff must interrupt customer service to prepare reports

- Repetitive manual data entry
- Risk of missing or delayed submissions
- High dependency on email communication
- No real-time visibility
- Risk of duplicate ordering
- No centralized tracking of request status
- Lack of automation or audit trail

The process is reactive, time-consuming, and prone to human error.

Proposed ERP Enhancement – Automated Minimum Order & Replenishment Engine

The proposed ERP replaces Excel-based reporting with an automated, system-driven replenishment workflow.

Branch/Store-Specific Minimum Order Configuration

The system allows the Branch Manager/Store Manager to define minimum order quantities for selected items at the branch/store level; however, configuring minimum levels is not mandatory for all items. Branch-wise configuration is essential for required items because item movement varies by location, customer preference, and regional demand patterns. Since the Branch Manager has the best operational understanding of local sales behaviour and fast-moving products, they are authorized to set thresholds that accurately reflect branch-specific consumption trends. This allows intelligent, localized inventory control.

Automatic Low-Stock Detection

When sales or stock transfers occur:

- Inventory updates in real time
- The system continuously checks minimum thresholds
- Once stock reaches the minimum level:
 - Item is automatically added to the Low-Stock List
 - No manual reporting required

Sales staff do not need to monitor or prepare Excel sheets.

Algorithm-Based Detection for Unconfigured Items (AI-Driven Reordering)

The system uses an AI-based algorithm to automatically detect items that are not properly configured for reordering or are running at risk of stock-out. This intelligent mechanism continuously monitors inventory movement across all branches and the main store, analyzing historical consumption patterns, seasonal demand, lead times, and current stock levels. Based on this analysis, the algorithm identifies items that either do not have predefined reorder levels or whose existing configuration is insufficient compared to actual usage trends.

When such unconfigured or under-configured items are detected, the system automatically adds them

to the **Less Item List** or **Reorder Recommendation Pool**. The concerned departments receive these recommendations along with suggested reorder quantities, preferred vendors, and expected delivery timelines. The suggestions are generated by comparing past purchase data, average consumption rates, and supplier performance.

This AI-driven approach reduces the dependency on manual monitoring, minimizes stock shortages, and ensures that frequently used items are always available. It also helps standardize the purchasing process by providing data-backed recommendations, improving decision-making, and optimizing inventory levels across all branches.

However, the accuracy and effectiveness of this algorithm will significantly improve with a minimum of three years of historical data.

With at least three years of transactional and consumption data, the system will be able to perform more reliable trend analysis, seasonal forecasting, and demand prediction, resulting in more precise low-stock identification and better purchase planning.

In summary, the algorithm acts as an intelligent assistant to the users, automatically detecting unconfigured items, recommending reorder actions, and supporting a more efficient and predictive procurement process.

AI Assistant ChatBot (Powered By SANDS Lab)

An AI-integrated chatbot will be implemented using OpenAI APIs. This chatbot will be connected to the purchase , sales or related datasets, enabling users to interact with the system through **natural language queries**.

The chatbot will analyze the available data and provide meaningful insights. For example, users can ask questions such as:

1. "Which product had the highest sales this month?"
2. "Which vendor supplied the most items in the last quarter?"
3. "What is the average purchase price trend for a specific product?"

The chatbot will process the dataset, interpret the query, and generate accurate and relevant outputs. This feature will enhance decision-making by allowing users to quickly extract insights without manually generating reports. However, the accuracy and effectiveness of this algorithm will significantly improve with a minimum of three years of historical data.

Review & Confirmation Workflow of Less Item List

Store/Branch Users can:

- Add items manually if required
- Remove incorrect entries
- Adjust quantities
- Update remarks

The Sales Manager reviews and confirms the list.

Once confirmed:

The request is automatically sent to the main store

- Items are tagged as Request Sent
- No email communication required.

Purchase Tracking & Status Visibility

Each item in the low-stock list includes lifecycle tracking:

- Request Sent
- Purchase Order Issued
- In Transit
- Arrived at Store
- Stock Updated

Branch/Store users can monitor status in real time.

This prevents:

- Duplicate ordering
- Panic reordering
- Miscommunication between departments

If a user tries to reorder an item already in process:

- The system displays a warning message
- Prevents accidental repetition

Cross-Branch Availability Visibility

Managers can view:

- Stock availability in central warehouse
- Stock in other branches

This enables smarter internal redistribution before external procurement.

Business Benefits

- Eliminates Excel reporting workload
- Reduces interruption to sales operations
- Fully automated low-stock detection
- Prevents duplicate purchases
- Centralized procurement coordination
- Real-time inventory transparency



- Faster replenishment cycle
- Improved stock planning
- Reduced emergency purchases
- Strong audit trail and governance

4. Less Item List Handling

Current Process

In the existing environment, the less item list is generated manually and handled through paper-based and fragmented workflows:

- The less item list will be received from the branches/store through email.
- The purchase team will print the less item list and handed over to the store team.
- Store staff physically verify each item against available stock and manually mark quantities on paper.
- The purchase team re-enters the entire list into the system.
- Purchase officers cross-check past transactions and purchase history item-by-item before raising RFQ.
- Searching historical transactions in the current system is slow and difficult, especially for previous accounting years.
- System performance issues and loading delays further increase processing time.
- Final RFQs are prepared and sent to suppliers by the Purchase Team or CTO.

This process is labor-intensive, error-prone, and heavily dependent on manual verification and duplicate data entry.

Proposed ERP Workflow

In the proposed ERP workflow, once branch managers confirm their individual Less Item Lists, the system automatically generates a consolidated multi-branch view for centralized store verification. This ensures that replenishment decisions are made based on complete organizational visibility rather than isolated branch requests.

Consolidated View Structure

The Store Manager receives a structured dashboard that aggregates confirmed requests from all branches in a single matrix-style report. The report includes:

- Item details
- Branch-wise required quantity
- Branch-wise available quantity
- Central store available quantity

- Central store required quantity
- Total available quantity across all locations
- Total required quantity across all locations

This consolidated visibility allows the Store Manager to make informed stock allocation and procurement decisions.

- Store Manager Verification Process
- The Store Manager performs a centralized review before sending to the purchase team
- Cross-checks actual central store inventory against system balances
- Verifies urgency of branch requirements
- Evaluates redistribution opportunities between branches
- Adjusts quantities if internal transfers can satisfy demand
- Confirms final consolidated requirement

Once verified:

- The consolidated request is digitally approved
- The system locks the request to prevent duplicate edits
- The approved list is automatically routed to the Purchase Department

System Controls and Safeguards

- All branch confirmations are timestamped
- Store-level adjustments are audit logged
- Changes require authorization
- Duplicate requests are automatically blocked
- Pending items remain visible in tracking dashboards
- Historical request records remain searchable

This ensures accountability and transparency across departments.

The consolidated less-item verification workflow transforms replenishment into a coordinated, organization-wide decision process. By providing the Store Manager with full multi-branch visibility before procurement, the ERP ensures optimal stock utilization, cost efficiency, and disciplined purchasing.

Internal Store-to-Branch Transfer Scenario

Before initiating external procurement, the ERP evaluates whether the central store can fulfill branch shortages.

If the store has sufficient available stock:

The Store Manager initiates a Store ☐ Branch stock transfer directly from the dashboard.

- Transfer quantities are logged in the system with:



- Item details
- Source location
- Destination branch
- Transfer quantity
- Timestamp
- Responsible user
- The transfer status is tracked (Initiated ☐ In Transit ☐ Received).
- Branch and store inventory balances update automatically in real time.
- The Less Item List is recalculated immediately:
 - Fulfilled quantities are removed
 - Remaining shortages are adjusted
- Only the net shortage is forwarded to the Purchase Department.

This ensures internal stock redistribution is prioritized before external purchasing.

The internal stock transfer workflow is explained in the Doc-003 V1.0

5. Purchase Review and Approval Workflow

Once the Less Item List has been verified and confirmed by the Store Manager, it is forwarded to the Purchase Department for detailed analysis and decision-making. The proposed ERP supports comprehensive visibility and workflow controls to ensure accurate, efficient, and auditable procurement.

5.1. Purchase Team Review

- **Dashboard Access:** The Purchase Manager or Purchase Officer can view the confirmed Less Item List in the ERP, which includes:
 - **Item details** – OEM Number, Brand, Company
 - Branch-wise and store-wise required quantities
 - Current stock levels
- **Transaction Analysis:** For each item based on the OEM number, the system displays recent activity including:
 - Branch-wise purchase transactions
 - Branch-wise sales transactions
 - Local purchases performed per branch
 - **Historical Data:** By default, the last 20 transactions are shown, with options to search for older transactions if required.
- **Decision Making:** Using this data, the purchase team decides:
 - Final procurement quantity per item

- Prioritization of purchase items based on demand, stock, and usage trends

- **Forwarding for Approval:** Once decisions are made, the list is submitted to the CTO for final review.

5.2. CTO Approval Purchase Request

The CTO receives a consolidated dashboard displaying each item with:

- Required quantity
- Potential vendors for procurement
- Delivery timelines
- Last purchase price
- Brand and OEM reference for requested items

CTO Capabilities:

- **Vendor Selection:** The CTO evaluates the list and decides which vendor(s) should receive the RFQ for each item.
- **Brand Adjustment:** Based on experience, the CTO can change the recommended brand for an item if suitable alternatives exist.
- **Quantity Confirmation:** The CTO confirms the quantity to be ordered per item, considering stock levels, branch needs, and procurement strategy.

Vendor Selection Method Using AI Algorithm modals

While the user identifies vendors, the system will suggest the best vendors based on the last 20 transactions, considering factors such as pricing and delivery accuracy. This evaluation will be performed using an AI-based algorithm, which helps the user select the most suitable vendor.

If the user chooses a vendor other than the one suggested by the system, the user must provide a comment explaining the reason for selecting a different vendor for that particular order. This action will be recorded in the system along with the user's name for tracking and audit purposes.

5.3. Request for Quotation (RFQ) Generation

Once approved:

The ERP generates multiple RFQ, mapping items to selected vendors automatically.

Vendor-Specific Links: The system can create a unique link for each RFQ and for each vendor, which can be sent directly to the respective vendor.

How this link works ?

The system will generate an RFQ (Request for Quotation) link for different vendors and send it to their registered email addresses. Each email will include the RFQ link along with a temporarily generated password for secure access.

When a vendor accesses the link using the specified password, they will be able to enter the price for each item and specify the expected delivery time. Once the vendor submits this information through the web form, the details will be automatically updated in the system.

At the same time, the submitted quotations from different vendors will be populated automatically in the system, allowing the user to compare them. The system will also highlight the best vendor based on pricing and delivery availability.

These links include:

- Item list (OEM, Part Number, Brand)
- Required Quantities
- Expected Delivery timelines
- Terms and conditions

Vendor Inputs: Vendors can provide the following information directly through the link:

- Available quantity
- Pricing per item
- Discounts (if applicable)
- Delivery timelines
- Terms and conditions

Export Option: For vendors who require offline documents, RFQ can be exported to Excel/PDF and sent manually.

How it works ?

When a user generates an offline RFQ document, the system will maintain a standard export format and assign a unique reference number to that document.

If the user sends the RFQ to the vendor without using the system link—such as through email, WhatsApp, or other communication methods—the system still supports the quotation update process.

When the vendor returns the quotation in the same Excel file with the required details filled in, the user can import that file into the system. The system will automatically map and update the submitted data against the corresponding RFQ using the unique reference number.

If the vendor sends the quotation through other formats, such as images or text messages, the user can retrieve the RFQ request from the system and manually update only the new values. This eliminates the need for multiple data entries and reduces the verification effort.

This process helps improve efficiency, maintain consistency, and minimize errors in quotation handling.

Tracking and Audit: Each RFQ link or export is logged in the ERP for tracking and audit purposes.



5.4. Business Benefits of RFQ Link and Offline RFQ

- Online RFQ Link
- Offline RFQ Document

Both methods are designed to ensure flexibility, accuracy, and efficiency in the procurement process. These options allow the organization to work with vendors who have different levels of technical capability and communication preferences.

Benefits of the RFQ Link (Online Quotation)

1. Real-Time Quotation Submission

Vendors can submit their prices and availability directly through a secure link. The data is instantly updated in the system, eliminating delays caused by manual entry.

2. Reduced Manual Data Entry

Since vendors enter their own pricing and delivery details, the system removes the need for users to manually input quotation data. This reduces:

- Human errors
- Duplicate entries
- Data inconsistencies

3. Faster Vendor Comparison

All vendor quotations are automatically populated into the system. This allows users to:

- Compare prices instantly
- Evaluate delivery timelines
- Identify the best vendor quickly

4. Improved Accuracy and Transparency

Each quotation is:

- Submitted directly by the vendor
- Time-stamped
- Linked to a specific RFQ

This ensures full transparency and auditability.

5. Secure Access

The RFQ link is:

- Password-protected
- Sent only to registered vendor email addresses





This ensures that only authorized vendors can access and submit quotations.

6. Automated Vendor Ranking

The system can automatically highlight the best vendor based on:

- Price
- Availability
- Past performance

This supports better and faster decision-making.

Benefits of Offline RFQ (Excel or Document-Based)

1. Flexibility for All Vendors

Some vendors may:

- Prefer email communication
- Have limited access to online systems
- Use WhatsApp or other channels

Offline RFQ allows these vendors to participate without technical barriers.

2. Standardized Format with Unique Reference

Each offline RFQ:

- Uses a standard export format
- Includes a unique reference number

This ensures that returned quotations can be correctly mapped to the original request.

3. Easy Import and Auto-Mapping

When vendors return the same Excel file:

- The user can import it into the system.
- The system automatically maps the updated data.

This eliminates the need for manual data entry.

4. Manual Update for Non-Standard Responses

If vendors send quotations via:

- Images
- PDF files
- WhatsApp messages
- Phone confirmations



The user can:

- Open the RFQ in the system
- Update only the changed values

This reduces multiple entries and simplifies verification.

5. Reduced Operational Effort

The offline RFQ process:

- Minimizes repeated data entry
- Reduces verification time
- Improves user productivity

6. Purchase Order Module

The Purchase Order Module manages the process of generating, evaluating, and sending purchase orders to vendors based on RFQ from vendors. The module ensures accurate comparison, proper documentation, and traceable communication with vendors.

6.1. Current Scenario

Currently, the CTO receives quotations from different vendors through various platforms such as email and WhatsApp. After collecting all the quotation details from these multiple sources, the CTO manually compiles the information into an Excel sheet.

All vendor pricing, availability, and related details are entered manually into the spreadsheet. Based on this data, the CTO prepares a price comparison table within Excel to evaluate the quotations.

The comparison of pricing and availability is performed manually, and the most suitable vendor is selected based on this evaluation. Once the decision is made, the CTO manually prepares and issues the Purchase Order (PO) to the selected vendor.

Due to the manual data entry, compilation, and comparison process, preparing a single Purchase Order becomes a time-consuming and labor-intensive task.

6.2. Proposed Purchase Order Generation

After the automated RFQ comparison (Refer : 5.4) is completed, the system suggests vendors based on its evaluation, and the CTO may either accept the system-suggested vendors or select vendors manually.

Once the vendors are finalized, the CTO can proceed to generate the purchase quotations or purchase orders.

In this process, the RFQ may contain multiple items. For example, if the RFQ includes 100 items:

- The first 10 items may be assigned to Vendor 1.

- The next 10 items may be assigned to Vendor 2.
- The remaining items can be distributed among other vendors based on price, availability, or strategic decisions.

The CTO has the ability to split and map the items to different vendors as required.

After this vendor-wise item allocation is completed, the CTO can click the "Generate Quotations" or "Generate Purchase Orders" button. The system will then automatically generate multiple quotations or purchase orders, each containing only the items assigned to the respective vendor.

These vendor-specific purchase orders can then be sent directly to each vendor's registered email address for confirmation via link or exported document.

6.3. Business Benefits

- **Efficient Vendor Comparison:** Automatic item-wise comparison ensures selection of optimal vendor(s) based on objective criteria.
- **Audit Trail:** All communication and response submissions are logged for compliance and future reference.

Reduced Manual Effort: Minimizes repetitive work of manually tracking responses and generating quotations.

Transparency: Provides clear governance with manual override controls and required justification.

Flexibility: Supports both link-based digital workflows and traditional manual entry for hybrid operations.

Faster Turnaround: Streamlines the quotation process, reducing procurement cycle time.

7. Purchase Verification Note

Current Scenario

In the current scenario, once vendors send invoices, the purchase team manually re-enters invoice items into the system to generate purchase verification note. This duplicate data entry consumes time, increases the risk of human error, and reduces visibility into quotation tracking.

In the proposed ERP, purchase order creation is directly integrated with the quotation module, eliminating repetitive manual work and improving traceability.

Proposed Workflow

7.1. Quotation/Invoice-to-PVN (Purchase Verification Note) Conversion

- Approved vendor quotations can be directly converted into PVN.
- The system automatically carries forward item details, quantities, pricing, and vendor information.
- All critical item details are automatically carried forward:
 - Pcode



- Alias
- OEM/Manufacturer part number
- Brand
- Part name
- Vehicle model
- Engine number
- Engine CC
- Position / Fitment
- Size / Specification
- Comments / Special instructions
- Quantity
- Unit of measure
- Pricing

7.2. Verification Stage

- After conversion, the purchase team reviews the generated list.
- Users can:
 - Cross-check invoice items
 - Modify quantities or values if required
 - Add remarks explaining changes
- All edits are logged for audit tracking.

7.3. Invoice Matching Control

- If an item exists in the quotation but is missing in the vendor invoice:
 - The system flags the discrepancy
 - User confirmation is required
 - Status remains Pending until resolved
 - If all quotation items are invoiced and delivered:
- The quotation status automatically changes to Completed

Some use case solution

If the user identifies an item in the invoice that is not present in the quotation, there are two possible scenarios:

Case 1: Item received from a previous quotation

The item may belong to a previous pending quotation from the same vendor. In this case, the system will provide an option to view the vendor's previous pending purchase orders or quotations.

- If the pending item is found to match the received item in the invoice, the user can select the corresponding PO item and update the record.
- The user will also have the facility to update the status of the purchase order, if required.

Case 2: Wrong item received

- If the item received is incorrect or was mistakenly dispatched by the vendor:
- The user can identify the item using the OEM number and map it to the correct PCode.
- The item can then be added to the Purchase Verification Note (PVN).
- If the item does not have a PCode:
- The user can request a new PCode from the CTO by submitting the item details.
- The item will be marked and highlighted as a Non-PCode item.
- The PVN will be created with this status until the PCode is assigned.

7.4. Partial Delivery Handling

- Items delivered partially remain tracked individually.
- The system maintains visibility of:
 - Ordered items
 - Delivered items
 - Pending items

7.5. Item-Level Visibility in PVN

Each PVN includes a detailed item list for operational clarity:

- PCode (internal identifier)
- Item name and alias
- OEM / Manufacturer part number
- Brand and part name
- Vehicle model, engine number, and engine CC
- Position / fitment
- Size / specifications
- Comments or special instructions
- Ordered quantity and unit of measure

This ensures that all users, from store to purchase and CTO, have full visibility of the items being procured, with reference to technical and operational details.

7.6. Business Benefits

The proposed ERP introduces full lifecycle tracking of purchase quotations and orders, giving management real-time operational visibility and long-term audit traceability.

Key benefits include:

- **Real-time purchase status monitoring**

All purchase quotations and PVNs can be tracked from creation through vendor confirmation, dispatch, partial delivery, and completion. Users can instantly view the current stage of each item and vendor commitment.

- **Prevention of duplicate reordering**

The system flags items already under active procurement, preventing accidental reordering of the same item while it is still in process.

- **Accurate delivery tracking**

Item-wise tracking ensures visibility of delivered, partially delivered, and pending quantities. This supports better warehouse planning and branch allocation.

- **Clear vendor performance visibility**

Delivery timelines, fulfilment accuracy, and pricing history are stored and analysable, enabling objective vendor performance evaluation.

- **Improved procurement planning**

Historical purchase data, pending orders, and demand trends help the purchase team make informed quantity and vendor decisions.

- **Permanent invoice attachment & retrieval**

Vendor invoices can be digitally attached to the corresponding purchase quotation/order inside the ERP. These documents remain permanently archived and can be retrieved at any time for audit, reconciliation, warranty reference, or dispute resolution.

- **Digital Document – Multiple Upload Facility**

The system will provide an option to upload multiple digital copies of documents received from the vendor. All related documents, such as invoices, delivery notes, quotations, and supporting files, can be uploaded and stored in the system against the respective document numbers.

Each uploaded file will be properly indexed and linked to its corresponding **Purchase Order (PO)**, invoice number, or **Purchase Verification Note (PVN)**. This ensures easy retrieval and structured record management.

This feature enables centralized document storage, allowing authorized users to access real reference documents at any time for verification, audit, or clarification purposes. It improves transparency, reduces dependency on physical files, and ensures that all purchase-related documents are securely maintained in one centralized system.

Operational Outcome

This workflow ensures that the purchase team can clearly see which items are:

- Ordered
- In transit
- Delivered
- Pending invoice
- Pending vendor action

The system acts as a live procurement dashboard rather than a static record system.

8. Reporting Module

The system will include a flexible and dynamic Reporting Module that allows users to generate reports based on the available data within the system.

Users will have the ability to create custom reports according to their specific operational and management requirements. In addition to custom reports, standard reports will be developed.

The reporting structure will be designed to support future enhancements, allowing reports to be customized or expanded as business needs evolve.

An AI-assisted reporting feature will also be implemented. This feature will enable users to generate reports using intelligent filtering and query options. Users will be able to apply multiple filters such as date range, branch, vendor, product, status, and other parameters to generate precise and meaningful reports.

The AI-assisted reporting module will help users quickly extract insights, reduce manual effort, and improve decision-making efficiency.

9. Management Review – Key Points (Current vs Proposed System)

9.1. Centralized Multi-Country Purchase Control

- All purchase activities will operate from a centralized location while serving multi-country and multi-branch operations.
- Vendor registration will support international suppliers under a unified master structure.
- Ensures standardization, pricing control, and reduced duplication across regions.

9.2. Vendor Communication Transformation

Current:

Communication handled via third-party tools (email, chat apps), causing scattered records and lack of traceability.

Proposed:

- Unique secure vendor links for purchase requests, quotations, and invoices.

- Integrated chatbot communication stored within the system.
- Full conversation history available for audit and future reference.
- Option for hybrid communication (system + external tools if required).
- Management Benefit: Controlled communication, transparency, and dispute-proof documentation.

9.3. Automated Less Item / Minimum Order Handling

Current:

Manual Excel reporting by branches email ,re-entry by purchase team ,high error risk and delays.

Proposed:

- Branch managers define minimum order quantity (optional per item).
- System auto-detects low stock during sales or transfers.
- Auto-generated Less Item List — no manual Excel required.
- Smart algorithm suggests reorder levels for items without fixed minimum quantity.
- Duplicate ordering prevention with real-time status tracking.

Management Benefit: Reduced human dependency, faster replenishment, and inventory accuracy.

9.4. Branch-Specific Inventory Intelligence

- Minimum quantity can be branch-wise due to local demand behavior.
- Branch managers apply location knowledge (fast-moving items differ by region).
- System respects operational flexibility while maintaining central oversight.

9.5. Store Verification Workflow

- Store team views consolidated branch shortage list.
- Available quantities auto-filled from live inventory.
- Store manager confirms required quantities.
- Stock transfer facility will be available with SOP.
- Adjusted list forwarded to purchase department.

Management Benefit: Internal stock optimization before external purchase.

9.6. Purchase Decision Dashboard

- Purchase team sees item movement (sales + purchases + branch local buying).
- Default last 20 transactions with deep search option.
- Data-driven quantity confirmation before escalation.

9.7. CTO Strategic Control Panel

- Required items displayed with vendor history.
- Delivery timelines, last purchase price, brand options per OEM.



- CTO may override vendor/brand based on experience.
- System logs decisions for governance.

9.8. Automated Multi-Vendor Purchase Order Generation

- System creates vendor-specific POs automatically.
- Unique link sharing or Excel export option available.
- Vendor interaction centralized and traceable.

9.9. Purchase Quotation Intelligence

- Vendors submit quotes through structured digital form.
- System auto-compares vendor responses item-wise.
- Smart vendor recommendation engine.
- Manual override allowed with mandatory justification.

Management Benefit: Balanced automation + human authority.

9.10. Global Price Master Governance

- System flags price differences vs existing item master.
- CTO receives alerts for price deviations.
- Approved price updates sync across all countries instantly.
- Maintains global pricing consistency.

9.11. Audit Trail & Accountability

Every confirmation step logged: branch store , purchase , CTO.

Chat communication archived.

Price changes recorded.

- Vendor selection reasoning stored.

Management Benefit: Compliance-ready operations.

9.12. Purchase Order Creation Governance

Current:

Purchase invoices are manually entered into the system, and purchase orders are recreated from scratch, increasing workload and risk of mismatch between quotations and invoices.

Proposed:

- Direct conversion of approved quotations into purchase orders
- System cross-verifies invoice vs quotation automatically
- Missing or extra items flagged for review



- Remarks option for justified deviations
- Partial delivery tracking maintained
- Quotation remains open until all items are fully delivered
- Real-time visibility of pending vs completed items
- Prevents accidental reordering of already committed stock

Management Benefit:

Full procurement lifecycle visibility, error reduction, and stronger delivery tracking discipline.

Thumb Rule for the Purchase Process

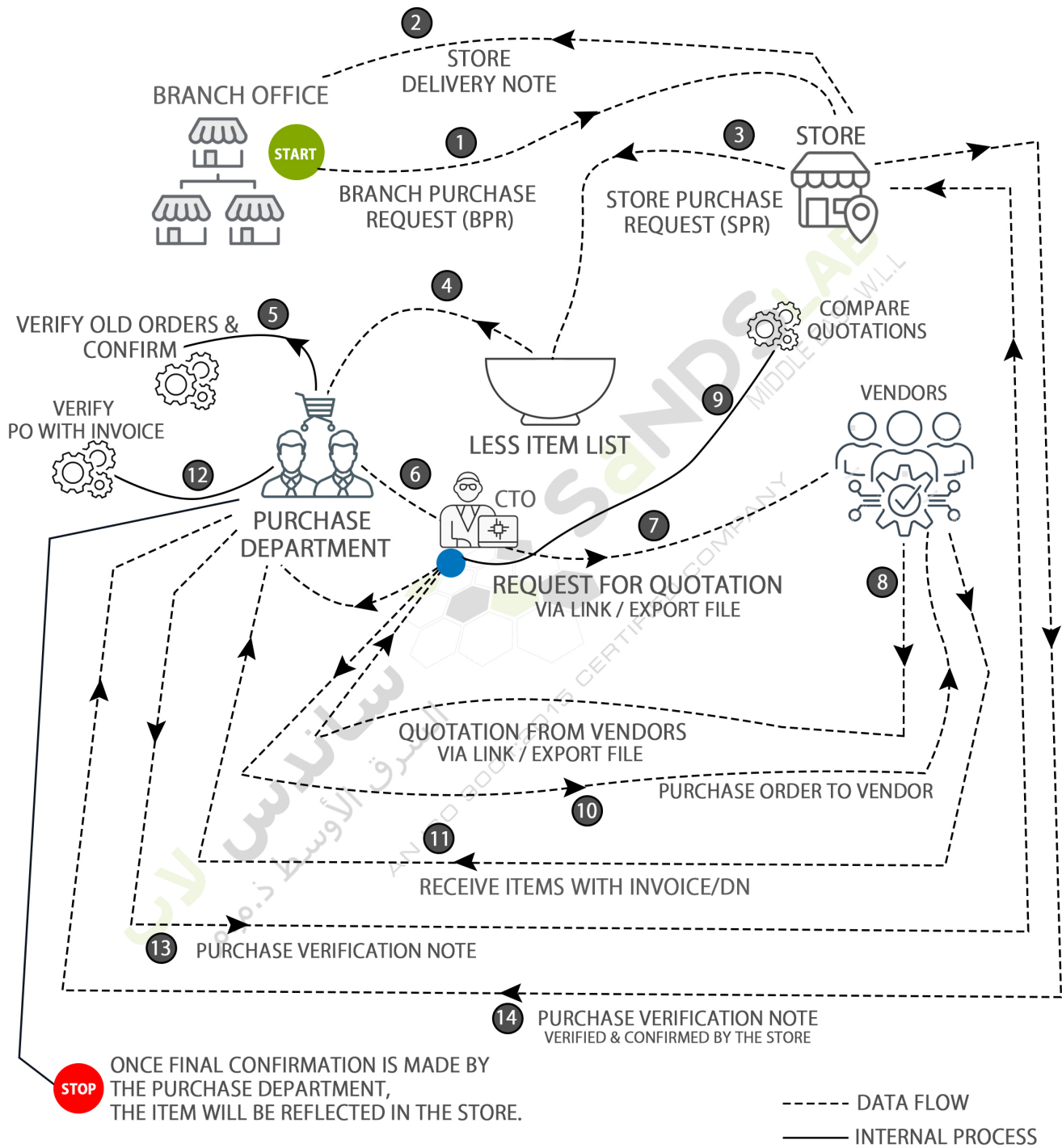
Simple Rule:

"If the store has the item, deliver it. If not, request, purchase, verify, and then update stock."

• Expanded Thumb Rule

1. Check stock first.
2. If available deliver to branch.
3. If not available raise purchase request.
4. Get quotations and issue PO.
5. Receive and verify items.
6. Confirm purchase.
7. Update stock and make it visible to all branches.

DATA FLOW DIAGRAM FOR PURCHASE





10. Data Flow Process for Purchase

Overview

This process explains how a stock request originating from a branch is handled through the main store, purchase department, CTO approval, vendor quotation, purchase order generation, goods receipt, and final stock update. The objective of this process is to ensure accurate procurement, controlled approvals, and proper stock visibility across all branches.

10.1. Step 1: Branch Requirement to Main Store

When a branch identifies that certain items are running low or out of stock, the branch manager creates a **stock request** in the system.

This request is automatically sent to the **Main Store**.

The request includes: (Details in the Page No : 8)

- Branch name
- Item details
- Required quantities
- Request date
- Requesting user

This ensures that all stock movements begin from a centralized process.

10.2. Step 2: Main Store Availability Check

Once the request reaches the Main Store, the Store Manager reviews the request and checks whether the requested items are available in the main store inventory.

If items are available:

1. The Store Manager assigns the request to a **Store Keeper**.
2. The Store Keeper collects the requested items.
3. The items are placed in the **designated branch pickup area**.
4. The system generates a **Store Delivery Note** for that branch.
5. The branch is notified that the items are ready for pickup.

This completes the process for items that are available in the main store. (Detailed Explanation in Doc-003 V1.0))

10.3. Step 3: Items Not Available – Creation of Store Purchase Request

If the requested items are **not available** or insufficient in the main store:

1. The Store Manager creates a Store Purchase Request.
2. These items are added to a centralized pool called the "Less Item List."



The Less Item List is a consolidated list of all items that need to be purchased due to shortages across branches.

10.4. Step 4: Purchase Department Receives Request

The Purchase Department accesses the Less Item List and reviews the pending purchase requests.

For each item, the purchase team identifies:

- Which branch requested the item
- Total quantity required
- Priority level

The purchase process then begins based on these details.

10.5. Step 5: Purchase Pattern Analysis

Before initiating procurement, the Purchase Department analyzes historical data, including:

- Previous purchase orders
- Vendor history
- Brand preferences
- Price trends
- Consumption patterns

This helps the purchase team:

- Select the right vendors
- Avoid overpricing
- Maintain consistency in brand or quality

After analysis, the request is forwarded for approval.

10.6. Step 6: Submission to CTO for Approval

The Purchase Department submits the analyzed purchase request to the CTO for approval.

The CTO reviews:

- Required items
- Quantities
- Historical data
- Budget considerations

Once satisfied, the CTO proceeds with the quotation process.

10.7. Step 7: Quotation Request to Multiple Vendors

The CTO selects the items and chooses one or more vendors from the system.

The system automatically:

- Generates quotation requests
- Sends them to vendors via email or secure link

Each vendor receives:

- Item list
- Required quantities
- Response link
- Attached PDF request

This enables competitive pricing.

10.8. Step 8: Vendor Quotation Submission

Vendors respond by submitting:

- Unit price
- Available quantity
- Delivery time

This information is submitted through the provided link or via email.

The system records all quotations automatically.

10.9. Step 9: Vendor Evaluation and Selection

The CTO reviews all submitted quotations using the system's comparison interface.

The evaluation is based on:

- Best price
- Product availability
- Delivery timeline
- Vendor performance history

After analysis, the CTO selects the most suitable vendor.

10.10. Step 10: Purchase Order (PO) Issuance

Once the vendor is selected:

- The system generates a Purchase Order (PO) automatically.
- The PO is sent to the selected vendor via:
 - Email, or
 - Secure system link





The PO includes:

- Item details
- Quantities
- Agreed prices
- Delivery terms
- Payment terms

10.11. Step 11: Vendor Shipment

After confirming the PO:

- The vendor prepares the shipment.
- The items are delivered to the main store.
- The shipment includes:
 - Invoice
 - Delivery Note

10.12. Step 12: Purchase Department Invoice Verification

Upon receipt of the shipment:

- The Purchase Department compares:
 - Invoice
 - Quotation
 - Purchase Order
- If all details match, the department generates a:

Purchase Verification Note

This note confirms that the documents are correct and moves to the Main Store for physical verification.

10.13. Step 13: Main Store Physical Verification

The Main Store receives the Purchase Verification Note and performs a physical inspection.

The store keeper:

- Opens the shipment
- Checks items against the invoice
- Verifies quantities and condition

If discrepancies are found:

- The system notifies the Purchase Department.

If everything is correct:

- The store manager confirms the verification.

10.14. Step 14: Final Purchase Confirmation

After the Main Store confirmation:

- The Purchase Verification Note is sent back to the Purchase Department.
- The Purchase Department makes any necessary adjustments based on:
 - Short deliveries
 - Damaged items
 - Quantity differences

The department confirms the purchase by approving the verification in the system.

10.15. Step 15: Stock Update and Visibility

Once the purchase is confirmed:

- The system updates the stock in the Main Store.
- The newly received items become:
 - Visible in the main store inventory
 - Available for all branches
- Branches can now:
 - View updated quantities
 - Request the items for replenishment

This completes the purchase data flow process.

End of Session